1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC shows heterogeneity in BCG (intravesical immunotherapy) benefit—some tumors with dense immune infiltration nonetheless recur, while others with low apparent local immunity respond; why do urine epigenetic signals and systemic mutational load modulate this discordance? 2) Tier 1 paths used: I built on (a) urine DNA methylation biomarkers as a non■invasive readout of tumor epigenetic programs that can suppress immune visibility (Core Path 1) and (b) bTMB as a systemic, non■invasive proxy for neoantigen burden that can drive adaptive immune activation (Core Path 3). I included TIL density (imaging feature) to capture local immune context (ties conceptually to Core Path 2/Immunoscore). 3) Tier 2 supporting evidence: TP53 and ER status can shape both mutation burden and epigenetic landscapes (modifying methylation and TMB); Immunoscore concepts validate that spatial immune context matters; TMB (tissue) and stool/urine methylation panels (SDC2/TFPI2) support cross■tissue epigenetic biomarker translatability. 4) Why therapy_response (BCG): the most actionable clinical problem in NMIBC is predicting/optimizing response to BCG; available non■invasive readouts (urine methylation, bTMB) are well suited as pre■treatment stratifiers for intravesical therapy. 5) Deliberate exclusions: I excluded proposing specific antigen■presentation genes or additional molecular features in the core hypothesis to keep focus to exactly one imaging metric + ≤2 molecular anchors; Tier■2 entities (TP53, ER status, Immunoscore, tissue TMB, SDC2/TFPI2) are listed as secondary_modifiers for follow■up mechanistic testing rather than included in the primary claim?

Working Hypothesis

In NMIBC, high intratumoral TIL density does not predict favorable response to BCG when pre-treatment urine DNA methylation signatures indicate tumor immune-visibility silencing, whereas elevated systemic neoantigen burden (bTMB) can counteract this silencing by promoting neoantigen-driven T cell activation. Therefore, pre-treatment integration of TIL density, urine methylation biomarkers, and bTMB will stratify NMIBC patients by likelihood of BCG response, guiding decisions toward adjunctive immunotherapy for those unlikely to respond.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	urine DNA methylation biomarkers, bTMB
Primary endpoint	therapy_response (BCG)

2. Why This Gene — Not Others

Primary hypothesis gene(s)

urine DNA methylation biomarkers: urine DNA methylation biomarkers as a non■invasive readout of tumor epigenetic programs that can suppress immune visibility (Core Path 1) and (b) bTMB as a systemic, non■invasive proxy for neoantigen burden that can drive adaptive immune activation (Core Path 3).

bTMB: bTMB as a systemic, non■invasive proxy for neoantigen burden that can drive adaptive immune activation (Core Path 3).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ER status	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: are recognized factors that may shape methylation landscapes or TMB and will be used for exploratory analyses rather than core hypothesis testing. Experimental approach (feasible methods):

Prediction 2: Assemble a prospective NMIBC cohort scheduled for BCG; collect pre-treatment tumor tissue for TIL density assessment (CD3/CD8 multiplex IHC or validated Immunoscore), urine samples for methylation profiling (targeted bisulfite sequencing panels or MeD-seq), and peripheral blood for bTMB (targeted NGS panel with standard TMB calculation).

Prediction 3: Define therapy_response to BCG using standardized NMIBC criteria (e.g., absence of recurrence/progression at defined follow-up intervals).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ER status: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.